

# Alex Dils

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## Education

University of California, Berkeley

May 2028

B.A. in **Computer Science** | GPA: 3.7/4.0

Coursework: Data Structures & Algorithms, Machine Structures, Discrete Mathematics & Probability, Multivariable Calculus

Activities: Machine Learning at Berkeley (ML@B), Internal & Research Committees

## Technical Skills

**Languages:** Python, JavaScript, TypeScript, Java, C, SQL; **ML:** PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, SimpleITK;

**Web/Data:** FastAPI, Node.js, React, PostgreSQL, Supabase, PostGIS; **Tools:** Docker, Slurm, Git, Vercel

## Experience

Stanford Canary CREST Research Scholar, Stanford Canary Center / Gevaert Lab

Jun 2026 – Present

- Built a **cNF volume-estimation pipeline** for NF1 cancer-risk surveillance, combining center-coordinate lesion localization, SAM v2 masks, Depth Pro surface reconstruction, and per-lesion  $\text{cm}^3$  integration.
- Generated synthetic RGB/depth lesions on Human Skin Repository 3D surfaces using physics-based protrusions and diffusion inpainting; fine-tuned Depth Pro, reducing depth-estimation AbsRel by **52.6%** and reaching  $R^2 = 0.97$ .

Machine Learning Engineer, Logitech

Feb 2026 – Jun 2026

- Built a **VLM evaluation harness** for meeting-room readiness and participant classification on Logitech's Rally Board 65, measuring task accuracy and inference latency across 55 models.
- Ran model and prompt ablations and identified Qwen3.5 4B as the leading deployment candidate for the target hardware and task.

Research Intern (part-time), Stanford Center for Biomedical Informatics Research

Jun 2022 – Jun 2026

- Developed domain-robust skin-lesion segmentation methods and finite-element lung-motion augmentations that **improved SoTA tumor-segmentation Dice by >0.05**.
- Drafted sections of a Digital Twin manuscript and its journal submission cover letter; the research was **featured on Stanford Medicine's homepage**.

## Projects

Openleaf | alex-dils.com/openleaf

2026

- Built a local macOS research workspace primarily in **JavaScript**, unifying LaTeX, PDF, Python, GitHub, SSH, and coding agents.

BsCode | alex-dils.com/bscode

2026

- Built a desktop primarily in **JavaScript** for coordinating Codex, Claude, and shell agents across local and remote workspaces.

LanternTrace Explorer | alex-dils.com/lanterntrace

2026

- Built a physics-based explorer to track and forecast spotted lanternfly spread across the U.S. East Coast.

## Publications

\* indicates first authorship.

- **\*Synthesizing Longitudinal Cutaneous Neurofibroma Imaging for Digital-Twin Burden Quantification.** MICCAI MWM 2026. Christoph Sadée, **Alex Dils**, Krish Sangani, Lillian Rubino, Alexander J Ferencick, Varun Wadhwa, et al.
- **\*Eye For An Eye: A Deep-Learning and Analytical Method to Spatializing Stereoscopic Images.** National High School Journal of Science, 2025. Jake Yoshinaka, **Alex Dils**, and Leo McDonnell.
- **\*Addressing Bias and Confounders in AI-based Image Diagnosis — A Study of 117,610 Skin Lesions.** Manuscript under revision at *Cell*, 2025. Christoph Sadée, Serena Zhang, **Alex Dils**, Jared Bitz, Stefano Testa, Zhuo Ran Cai, et al.
- **Integrating Mechanistic Knowledge into Deep Learning for Improved Cancer Detection.** FEniCS Conference Proceedings, 2024. Christoph Sadée, Katherine Hartmann, **Alex Dils**, Ianto Xi, Francisco Carrillo-Perez, et al.